Akshat Semwal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.45
Rishiram Shikshan Sansthan	Class XII (CBSE)	2023	95.40%
HIM Christian Academy	Class X (ICSE)	2021	89.30%

SKILLS

- **Data & Analytics:** SQL (MySQL: CTEs, window functions, cohort and RFM analysis), Python (Pandas), Excel (Power Query), Power BI
- **Product & Business Analysis:** User stories and Gherkin acceptance criteria, Requirements Traceability Matrix, BPMN (MS Visio), Agile roadmapping, Gantt planning

PROJECTS

OmniPulse Analytics: E-Commerce & SaaS Intelligence Engine | MySQL, Python | Github

- **Modeled a dimensional MySQL database** (3 dimension, 2 fact tables) holding **14,000+** transactional and billing records (15,500+ total rows), consolidating revenue, retention, and churn analysis into a unified schema using Python (Faker, Pandas).
- **Computed MoM revenue growth** and 3-month moving averages via multi-level CTEs and window functions (LAG, dynamic rolling frames), pinpointing peak net revenue of $52,400 at a **14.8%** MoM growth rate.
- **Segmented customer lifecycle** with RFM scoring (NTILE) to reveal Champions (6.2% of users) drive **28.4%** of total revenue; engineered a Month 0-6 cohort retention matrix (PERIOD_DIFF) tracking an **18.5%** Month-1 retention rate.
- **Analyzed MRR, churn, and invoice failure rates** by plan tier, isolating Enterprise Monthly as the primary source of revenue leakage (**11.4%** churn); instantiated 3 production SQL views to feed executive BI reporting layers.

Enterprise CRM Implementation Framework (Capstone, simulated engagement) | Microsoft Business Analyst Certificate

- **Synthesized** high-level business goals (EPICs) into a backlog of user stories with testable Gherkin acceptance criteria, by decomposing requirements until each had a clear definition of done.
- **Created** a Requirements Traceability Matrix achieving **100%** alignment between functional requirements, non-functional benchmarks (**500+** concurrent users), sprint tasks and test cases.
- **Translated** an **8-sprint** roadmap into a critical-path Gantt chart, balancing resource demand across UX, Dev and QA.

Autonomous UAV Systems | Research Intern, IIT Bombay | May 2025 - July 2025

- **Researched** trade-offs among flight controllers, sensor redundancy and ESC protocols across components, producing a documented comparison to guide hardware assumptions.
- **Built** a Docker/PX4/Gazebo simulation environment that validated hardware assumptions before prototyping, reducing verification time; **debugged** a real-time GStreamer pipeline to restore continuous sensor feedback.

RELEVANT COURSES

Microsoft Business Analyst Professional Certificate | *Coursera*

- **Core Skills & Competencies:** Comprehensive 6-course program covering Business Analysis frameworks, advanced data manipulation in Excel, and end-to-end project delivery.
- **Requirements & Modeling:** Proficient in elicitation techniques, stakeholder communication, and visual workflow/data architecture mapping using standard BPMN notation in MS Visio.
- **Data & Analytics:** Experienced in cleaning, transforming, and modeling large-scale datasets using Excel (Power Query, dynamic dashboards) and Power Platform to build decision-support visualizations.
- **Capstone Execution:** Applied structural frameworks to evaluate, scope, and validate real-world business scenarios from project initiation to final delivery.

Innovation Management, Lean Startup, Communication Skills, Construction Management

Introduction to Programming, Digital Fabrication, Fabrication Lab, Engineering Drawing

POSITIONS OF RESPONSIBILITY

Speaker and Content Curation Head | EML(Extra Mural Lectures)

- Leading content strategy to launch a high-engagement digital 'Hostel Series' spotlighting legendary Indian scientists alongside dedicated segments on prestigious national award laureates.

Coordinator (Collaboration & PR) | EML(Extra Mural Lectures)

- Driving strategic public relations and audience outreach campaigns to scale event footprint and brand visibility.

Assistant Manager (Operations) | E-Cell

- Directing end-to-end logistics, vendor relations, and startup stall allocations for flagship events E-Merge and E-Summit.

EXTRACURRICULAR

- **Adventure Sports:** Completed a bungee jump at Rishikesh and successfully navigated a 26 km white-water river rafting expedition through challenging rapids.
- **Trekking & Mountaineering:** Completed multiple multi-day high-altitude Himalayan expeditions, demonstrating physical endurance and resilience across challenging wilderness terrains.
- **Sports & Athletics:** Actively competed in multiple inter-hostel and department tournaments across the institute, representing teams in cricket and badminton.